Semiconductor Physics And Computational Methods

Sem 2 | Watermarked study resource | Generated from local SRM dataset

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[Page 1]
Reg. No.
Time: 3 hours Max. Marks: 75
Marks BL CO PO
1. 1 1 1 1
1 12. i i

3. i i i i
i i i4. i
i i 2 15.
i i 2 16.
i 2 I7.
21 1 18.
i i 3 39.
31 1 3

Note:
(i) Part - A should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall
invigilator at the end of 40th minute.
(ii) Part - B & Part - C should be answered in answer booklet.
PART - A (20 x 1 = 20 Marks)
Answer ALL Questions
with an increase in temperature.
B) Increases
Which of the following represents a indirect bandgap semiconductor?
A)GaAs B) GaAs and Si
C)Ge D)Si
B.TECH / M.TECH. (Integrated) DEGREE EXAMINATION, DECEMBER 2024
Frist Semester
21PYB102J - SEMICONDUCTOR PHYSICS AND COMPUTATIONAL METHODS
(For the candidates admitted during the academic year 2024-2025)
B) Psc/IscVSc
D)Pm*IscVsc
Which fundamental equation governs the quantum mechanical treatment of electrons?
A)Maxwell’s equations B) Schrodinger equation
C) Newton's second law D) Boltzmann equation
Consider a one-dimensional lattice of length a. The length of the unit cell in the
reciprocal
lattice is .
A)l/fl B)7c/<7
C)47r/a D)a
The quantization of lattice/atomic vibrations is called as -—
A) Photon B) Exciton
C)Phonon D) Magnon
B)hv = Eg/C
D)hv>E„ o
The Fermi level of n-type semiconductor
A) Becomes zero
C) Remains unchanged D) Decreases
The rectifying metal - semiconductor junction is also called as
A) Ohmic junction B) Schottky junction

C) Conducting junction D) PN junction
In an intrinsic semiconductor at 0 K, the Fermi level (or chemical potential) is located at
1
A) Close to the valence band B) Close to the middle of the band gap
C) Close to the conduction band D)In the conduction band
The random motion of holes and free electrons due to thermal agitation is called
A) Diffusion B) Drift
C) Pressure D) Ionisation
The condition for photon absorption is
A)hv = C/Es
C)hv<Ee o
10. Which is the correct formula for fill factor?
A) Pm^sc^sc
C)Pm/IscVoc
THE HELPERS

1 1 3 3
31 1 3
11 1 4
1 41 1
4 11 115.
i 41 1volts.
1 1 5 1to mention about nanoscale17.
i 1 5 1
1 51 1
1 1 5 1
Marks BL CO PO
21 a.(i)What are the assumptions of classical free electron theory and its key successes? 8
3 1 1
8 2 1 1
(ii) Write a comparison between localized and delocalized wave functions?
8 4 2 1
(OR)

B) Hot point probe
D) linear
22 a. Analyze the P-N junction diode mechanism under the forward and reverse bias with a proper diagram.
PART - B (5 x 8 = 40 Marks)
Answer ALL the Questions
B)Prof. Richard Feynman
D)Prof. Einstein
dimensional structure is a delta function.
B)One

D) Three
(ii) Calculate the probability of non-occupancy for the energy level which lies 0.01 eV above the Fermi energy level at 27 °C. (KB = 8.625 x 10'5 eV/K)
(OR)
b. (i) Discuss the classification of electronic materials based on band gap and resistivity.
B) Solid
D) Semisolid
16. The potential barrier for Si in p-n junction is approximately
A) 0.3 B)0.5
C)0.7 D)0.9
“There is plenty of room at the bottom” is the phrase by
physics.
A) Prof. C V Raman
C)Prof. Schrodinger
18. Density of states for
A) Zero
C)Two
19. An example for two dimensional material is
A)Nanowire B) Nanosheet
C) Nanoparticle D) Quantum dot
20. In chemical vapour deposition, the precursors are introduced to the reaction chamber in the
state.
A) Liquid
C) Gaseous
11. The Fermi's golden rule helps to identify the
A) Velocity of light B) Momentum of electrons
C) Absorption energy of electron D) Transition rate per unit volume
12. Which process of an electron-hole pair is responsible for emitting light?
A) Generation B) Recombination
C) Ionisation D) Diffusion
13. method is the easiet method to investigate whether the given sample is n-type
or p-type semiconductor.
A) Two probe
C)Four probe
14. TCAD stands for
A) Topography Computational Aided Design B) Tribilogy Computer-Aided Design
C) Technology Computer-Aided Design D) Transmission Computer-Aided Design
method is most suitable for measurement of very high resistivity.
A) Two probe B) Four probe linear
C) Van der Pauw D) Three probe linear
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8 4 2 1

8 4 3 3
8 4 3 3
8 3 4 1
8 3 4 1
8 2 5 1
8 2 5 1
Marks BL CO PO
15 426
15 327
Explain the special properties of carbon nanotubes.
******

24 a. What is Hall effect? Derive Hall coefficient. Relate the Hall coefficient with Hall voltage.
25 a. Explain the construction and working of the physical vapour deposition technique with a
neat sketch.
23 a. Define the term density of states for photons. Derive the expression for the density of
states for photons.
b. Derive the continuity equation for electron an infinitesimal slice thickness of dx in a one-dimensional system.
Using the Bloch theorem, explain the Kronig-Penney model in detail with suitable
diagrams. Derive the expression for an electron in a periodic potential of a crystal with cases.
Analyse the construction and working of transmission electron microscopy with a neat sketch.
PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Questions
(OR)
b. Derive the Einstein coefficient expression for the ratio between spontaneous and stimulated emission.
(OR)
b. Discuss in detail the principle and working of the capacitance-voltage measurement technique.
(OR)
b. With a neat sketch analyze the working of atomic force microscopy.
i i
4 5
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College of Engineering and Technology
DEPARTMENT OF PHYSICS
B.Tech/M.Tech Integrated DEGREE EXAMINATION

Faculty Name: Dr. A. Senthil FacultyID: T239
Course Code: 21PYB102J
Course Name:Semiconductor Physics and Computational Methods
Date of Exam: 06/01/2024 Duration:3 Hours

PART—A (20x1=20 Marks)
Answer ALL Questions

1. (C)Sommerfeld
2. (B)0.5
3. (B)Insulator
4. (C) e 2X
5. (A)Intrinsic
6. (D)Light, Electric current.
7. (C)Movement of charge carriers in the presence of an applied electric field.
8. (B) Light Emitting Diode
9. (C)0.87 µm

10. (A)Number of atoms in excited state
11. (D)Fermi's Golden Rule
12. (A)Beam Propagation Method
13. (D)Vander Pauw method
14. (B)Diffusion potential
15. (B)V/I
16. (D)0.333 ohm m
17. (D)1-100 nm
18. (B)All three dimensions
19. (C)Gas
20. (B)AFM

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Course Code: 21PYB102J
Course Name:Semiconductor Physics and Computational Methods
Date of Exam: 06/01/2024 Duration:3 Hours

PART– B(5 x 8= 40 Marks)
Answer ALL Questions
Q-21 (a) Defination- Diagram-Importance of density of State - 2 marks
Definition: Density of States Z (E) dE is defined as the number of available electron states per unit volume in an energy interval (dE).

Derivation – 6 Marks

(OR)

[Page 7]
(b) (i) Explanation of Brillouin zone – 2 marks;

Diagram of Brillouin zone 1D,2D Lattice- 2 marks

(ii) Concep tof Eigen function,Eigen value and Eigen equations : 3marks ;
Examples: 1mark

In quantum mechanics, if we apply an operator on wave function then the same wave function is achieved back with a scalar value. In that scenario, the Wave function is called
Eigen function and the scalar value is called Eigenvalue. An equation that includes Eigen function, operator and Eigen value is called the Eigen equation.

No same function is achieved back after applying differential oper ator, (i.e.Sin and Cos functions are different). Where 3 is eigen value The samefunction is achieved back with a multiple of ‘3’ after applying the differential operator.

[Page 8]
Q.22 (a). Variation of Fermi level with temperature for N- type (4 marks) ;

Variation of Fermi level with temperature for P- type (4 marks)

(OR)

(b) Processes of diffusion involved in formation of a PN junction – 3

Continuity equation for transport of mobile charge carriers - 5 marks

Definition:1 mark

Equation with explanation: 4 marks

The contribution of the overall effect when drift, diffusion, and recombination occur simultaneously in a semiconductor material. The governing equation is called the continuity equation

The continuity equa tion describes a basic concept, namely that a change in carrier density over time is due to the difference between the incomingand outgoingflux of carriers plus the generation and minus the recombination.

Consider an infinitesimal slice with thickness dx l ocated at x. The number of electrons in the

slice may increase due to the net current flow into the slice and the net carrier generation

in
the slice (mainly four components contribution).

[Page 10]
Q.23 (a) Density of state of Photon : Explanation – 2 mark; Equation with explanation: 6 marks

(OR)

[Page 11]
(b) (i) Concept and explanation of Fermi Golden rule :3marks

Equation:2 marks

(ii) Determine the conversion efficiency of the solar cell
if short-circuitcurrent (I_{sc})=2.8 A, open- circuit voltage (V_{oc})=0.55V,
fillfactor(FF)=0.8,and inputpower(P_{in})=10W.
Answer: Equation- 1mark; Calculation and Answer -2marks

Conversion efficiency = 12.32 %

24. (a) Hall effect : Difinition of Hall effect with Diagram – 2 marks;
Relationship between Hall coefficient and Hall voltage derivation – 6 marks.

[Page 12]

(OR)

[Page 13]
(b) (i) Working principle and explanation of Hot probe :3marks ; Diagram: 1 mark

(ii) Monte Carlo method – Difinition 1` mark ;

Explanation of estimation of Pi value – 3 marks

[Page 14]

25. (a) Difinition and derivation of 2D and 1D – 6 marks (Each 3 mark s); 0D – explanation -
2 marks;

2D 1D
0D

[Page 15]

2D DOS

1D DOS

(OR)
25 (b) PVD – Diagram – 2 marks; Explanation – 6 Marks

 Alternative process to electroplating.

 The process is similar to chemical vapour deposition (CVD) except that the raw materials/precursors, i.e. the material that is going to be deposited starts out in solid form, whereas
in CVD, the precursors are introduced to the reaction chamber in the gaseous state.

Evaporation

During this stage, a target, consisting of the material to be deposited is bombarded by a high energy
source such as a beam of electrons or ions. This dislodges atoms from the surface of the target,
'vaporising' them.

2
1
22
�
�� Ek
Lm
dE
dn
��
22 ��
Am
dE
dk
dk
dn
dE
dn ��

[Page 16]

Transport

This process simply consists of the movement of 'vaporised' atoms from the target to the substrate to be
coated and will generally be a straight line affair.

Reaction

In some cases coatings will consist of metal oxides, nitrides, carbides and other such materials.
In these cases, the target will consist of the metal.
The atoms of metal will then react with the appropriate gas during the transport stage.
For the above examples, the reactive gases may be oxygen, nitrogen and methane.
In instances where the coating consists of the target material alone, this step would not be part of the
process

Deposition

This is the process of coating build up on the substrate surface.
Depending on the actual process, some reactions between target materials and the reactive gases may also
take place at the substrate surface simultaneously with the deposition process.

Fig. shows a schematic diagram of the principles behind one common PVD method.
The component that is to be coated is placed in a vacuum chamber. The coating material is evaporated by
intense heat from, for example, a tungsten filament.
An alternative method is to evaporate the coating material by a complex ion bombardment technique.
The coating is then formed by atoms of the coating material being deposited onto the surface of the
component being treated.

[Page 17]

PART—C (1x15=15Marks)

Answer ANYONE

26 (i) Derivation for an electron in a periodic field of 1D crystal lattice- Diagram with explanation – 2 marks ; Derivation – 10 marks

According to Kroning and Penney the electrons move in a periodic square well potential.

 This potential is produced by the positive ions (ionized atoms) in the lattice.

 It is not easy to solve Schrödinger’s equation with these potentials. So, Kroning and Penney
approximated these potentials inside the crystal to the shape of rectangular steps as shown in Fig.
(c). This model is called Kroning-Penney model of potentials.

(ii) Explanation of energy of electron under p 0 (3 marks)

[Page 18]

27 (i) Construction working principle of TEM - 4 marks

the electron gun, produces a stream of monochromatic electrons.

This stream is focused to a small, thin, coherent beam by the use of condenser lenses 1 and 2. The first lens (usually controlled by the "spot size knob") largely determines the "spot size"; the general size range of the final spot that strikes the sample.

The second lens (usually controlled by the "intensity or brightness knob" actually changes the size of the spot on the sample; changing it from a wide dispersed spot to a pinpoint beam.

The beam is restricted by the condenser aperture (usually user selectable), knocking out high angle electrons (those far from the optic axis, the dotted line down the center)

The beam strikes the specimen and parts of it are transmitted

This transmitted portion is focused by the objective lens into an image

The image is passed down the column through the projector lenses, being enlarged all the way.

The image strikes the phosphor image screen and light is generated, allowing the user to see the image

Comment on the interaction of electron beam with sasmple in SEM and TEM

– 6 marks

Specimen Interactions

TEM- Unscattered Electrons ; Elastically Scattered electrons ; Inelastically Scattered Electrons

SEM- Backscattered electrons ;Secondoryelectrons ; X-rays ;Auger electrons

(ii) Working concept of AFM – 4 marks ; Diagram – 1 mark

[Page 19]

 The AFM raster scans a sharp probe over the surface of a sample and measures the changes in
force between the probe tip and the sample.

Working Concept

 The physical parameter probe is a force resulting from different interactions.
 The origin of these interactions can be ionic repulsion, van der Waals, capillary, electrostatic and
magnetic forces, or elastic and plastic deformations.
 Thus, an AFM image is generated by recording the force changes as the probe (or sample) is scanned in the x, y and z directions.
 The sample is mounted on a piezoelectric scanner, which ensures three-dimensional positioning
with high resolution.
 The force is monitored by attaching the probe to a pliable cantilever, which acts as a spring, and
measuring the bending or "deflection" of the cantilever.
 The larger the cantilever deflection, the higher the force that will be experienced by the probe.
 Most instruments today use an optical method to measure the cantilever deflection with high
resolution; a laser beam is focused on the free end of the cantilever, and the position of the
reflected beam is detected by a photodiode.
 AFM cantilevers and probes are typically made of silicon or silicon nitride(nano) by micro
fabrication techniques.

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[Page 1]
b.i. What is a Light Emitting diode (LED)? Describe the principle, construction and working of
(8 Marks)
(4 MarkO
30. a. Explain the absorption and emissions processes with necessary theory and hence derive.the
relation between Einsteins coeflicients.
b.i. Describe the theory of Drudes ,nou!?llro 0".,."
conductivity.
ii. Mention anytwo applications of photovoltaic effect.
31. a. Describe the Linear and Vander Pauw Four Point Probe technique
measurements.
LED.
ii. Write a note on diffirsion and ddft current.
B.Tech. DEGREE EXAMINATION, NOYEMBER 2018
First Semester
18PYB103J - PITYSICS: SEMICONDUCTOR PITYSICS
@or the candidates admitted during the academic year 2018-2019)
Part - A should be answered in OMR sheet within first 45 minutes and OMR sheet should be

handed
b. What are the fundamental laws or alsoo?tionz Describe the principle, construction ,np working of UV Visible Spectrophotometer.
derive the expressionfor electrical
(10 Marks)
(2 Marks)
for eleckical
(Marks)
(8 Marks)
(8 Marks)
(4 Marks)
Note:
(i)
over to hall invigilator at the end of 45d minute.
(iD Part - B and Part - C should be answered in answer booklet
Time: Three Hours
(A) Collisiontime
(C) Wave number
PART-A(20x1=20Marks)
AnswerALL Questions
1. The ave.rage distance travelled by an electon between two successive collisions in the presence ofapplied field is called
@) Mean free path
(D) Drift velocity
@) Zener diode
@) Transistor
Max. Marks: 100
and holes is the
direc{ly emit a
- hole pairs, some of the
32. a.i. What are Carbon nanotubes (CNT)? Mention the properties of CNTs.
ii ii. Describe the fabrication of CNT's by Physical Vapor Deposition @VD).
Describe the principle, construction and working of Scanning Electron Microscope (SEM).
The band gap is called _, if the crystal momentum of electrons
same in both the conduction band and the valence band; an electron can
photon.
(A) Direct (B) Indirect
Write a note on Heterojunctions.
(oR)
Z.
3.
4.
b.i.
ii. (C) Crystalline
temperature above 0K.

(A) Valence level
(C) Conduction level
(A) Restmass
(C) Zero mass
(A) Light Emitting diode
(C) Rectifier
(A) gain
(C) amplification
impurity?
(A) N-qryesemiconductor
(C) P-typesemiconductor
(B) Fermi level
(D) Density of states
@) Effectivemass
@) Accelerated mass
is aPN Junction, which is forward biased?
(D) Noncrystalliae
is the state at which the probability of electron occupation is t/z at any
**.*tl.*
When an electron in a periodic potential is acceleiated relative to the lattice in an
electric
field or magnetic field, then the mass of the electon is called the
6. When light impinges upon a semiconductor to create electron
cariers are collected at the contact which leads to
(B) photocunent
(D) biasing
7. Which type of material is obtained when intrinsic semiconductor is doped with pentavalent
(B)
(D)
Extrinsic semiconductor
Insulator

8. is the process of radiative recombination of electron-hole pairs 18. In a quantum wire,
the material size is reduced
(A) In three directions
(C) In one direction
Charge
Potential
effective mass of hole.
9. Thp spectral region, where the material changes from being relatively transparent to
strongly
absorbing is known as
(A) Absorption edge

(C) Valence edge
10. According to Drudes theory, the velocities of electrons are assumed to have velocity given by kinetic theory of gases.
(A) Rootmeansqure (B) Drift
(C) Instantaneous (D) Unifomt
is the creation ofvoltage and electic current tn
19. In CVD chamber, the precursors are introduced to the reaction chamber in the state.
(A) Liquid
(C) Semisolid
20. The physical paxameter that is
created by electron bombardment.
(A) Photoluminescence
(C) Elecftoluminescence
(A) Acousto- optics
(C) Eleclrolysis
@) Cathodoluminescence
@) Anodoluminescence
(B) Conduction edge
(D) Annihilation edge
(B) In two directions
@) Infinitely
(B) Solid
(D) Gaseous
probed in AFM resulting from different interactions is
Force
Field
(10 Marks)
1.9 eV. Calculate the velocity of the electron at the Fermi
(2 Marks)
(4 Marks)
17I\IF'I18PYB1O}I
(B)
(D)
(A)
(c)
11.
13. For determining the resistivity of a serniconductor, the diameter of contacts between the
proQe and the semiconductor should be , the gap between the probes.
PART-B(5x4=20Marks)
ArswerANYFM Questions
21. Write a note on Enerry bands in solids.
22. Describe the nonequilibrium propeties of carriers.
23. Whatis a PN Junction? Explain the biasing concgpt in PN Junction.

25. Describe the optical absorption and recombination process,
26. Write a note on I-V charaoteristics of a diode
27. Describe the powder method of X-ray diffraction.
PART-C(5x12:60Marks)
Answer ALL Questions
28.a.i. What is Density of states? Derive an expression for density of states for a semiconducting
. (B) Photovoltaics
(D) Electrophorsis
a material .rFon exposure
to light and is a physical and chemical phenomenon
12. Instimulated emissiorq the states at which the life time of atoms is extended is
(A) Metastable state (B) Stable state
(A) Smallerthan
(C) Equal to
t4-
1s. A
(A) Surface area
(C) Voltrme

t6-
17.
where the applied voltage is varied, and the capacitance is measured and plotted as a function
ofvoltage.
(A) Capactive - voltage profiling
(C) Voltage profrling
is a method of determining quickly whether a semiconductor sample is n
(B) Greater than
@) Double
is a technique for chmacterizing semiconductor materials and devices,
@) Currentprofiling
@) Baising
@) Surface charge
(D) Force
17I\[Fl1EPYBlO}I
material.
ii. The Fermi level for potassium is
level.
(oR)
b. Describe the behavior of electron in a periodic potential and hence explain the Kronig Penny
Model in detail with the cases.
29.a,i. What is an extrinsic semiconductor? Describe the variation of Fermi lev,el with carrier

concenfiation and ternperature in an N-Type semiconductor. (8 Marks)
ii. Determine the position of the Fenni Level in an instinsic semiconductor from the centre of
forbidden gap at room temperature, if the effective mass of an electron is equal to twice the
(negative) type or p (positive) type.
(A) Eleckolysis @) Hot point probe
(C) Rectification (D) Hydrogenation
law states that, when a beam of monochromatic light passes tluough an
absorbing medium, the rate of decrease in intensify with the thickness of the medium, is
proportional to the intensity of light.
(A) Larnbert's @) Beer's
(C) Photoelectric @) Snell's
Nanoparticles are special mainly because of their
Page 3 of ,l
(oR)

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